SECTION 26 23 00 - LOW-VOLTAGE SWITCHGEAR

Project Alpha - Data Center Building DC-1, Electrical Room ER-1. Issued for Construction, Rev 2.

PART 1 - GENERAL

1.1 SUMMARY. A. Section includes metal-enclosed, drawout low-voltage power circuit breaker switchgear SWG-1 serving the UPS and mechanical loads.

1.2 SUBMITTALS. A. Product data, shop drawings with dimensions, short-circuit ratings, seismic certificates and factory test reports.

1.5 WARRANTY. A. Manufacturer's warranty shall cover parts and labor for not less than 24 months from the date of Substantial Completion.

1.6 DELIVERY. A. Switchgear shall be delivered to site not later than 30 weeks after receipt of approved submittals.

PART 2 - PRODUCTS

2.1 BASIS OF DESIGN. A. Manufacturers' standard products meeting every requirement of this Section. Substitutions per Section 01 25 00.

2.2 RATINGS. A. System voltage: 480Y/277 V, 3-phase, 4-wire, 60 Hz.

2.2.B Main horizontal bus continuous current rating shall be not less than 4000 A.

2.2.C Short-circuit current rating shall be not less than 65 kA RMS symmetrical at 480 V. Equipment shall be fully rated; series ratings are not acceptable.

2.3 CONSTRUCTION. A. Bus: silver-plated copper throughout, including main, riser and neutral bus.

2.3.B Neutral bus shall be 100 percent rated.

2.3.C Ground bus: copper, extending the full length of the lineup.

2.3.D Enclosure: NEMA 1 indoor, UL 1558 listed and labeled.

2.3.E The overall lineup length shall not exceed 240 inches to fit Electrical Room ER-1.

2.3.F Finish: ANSI 61 light gray, manufacturer's standard powder coat.

2.4 CIRCUIT BREAKERS. A. Drawout, electrically operated low-voltage power circuit breakers listed to UL 1066.

2.4.B Trip units: microprocessor-based with long-time, short-time, instantaneous and ground-fault (LSIG) functions.

2.4.C Arc energy reduction: provide an energy-reducing maintenance switch per NEC 240.87 on every breaker rated 1200 A or higher.

2.5 METERING AND COMMUNICATIONS. A. Main power meter shall be revenue-grade, accuracy class 0.2 per ANSI C12.20.

2.5.B Communications: native Modbus TCP and BACnet/IP to the Electrical Power Monitoring System.

2.6 SURGE PROTECTION. A. Integral surge protective device listed to UL 1449 4th Edition, with surge current rating of not less than 200 kA per phase.

2.7 SEISMIC. A. Switchgear shall be seismically qualified per IBC 2021 and ICC-ES AC156 for the project site (SDS = 1.0 g), with certificate of compliance.

PART 3 - EXECUTION

3.2 FIELD QUALITY CONTROL. A. Factory witness testing: provide factory witness testing for up to two Owner representatives prior to shipment.

3.3 STARTUP. A. Manufacturer's field service engineer shall perform startup, testing and commissioning support.

END OF SECTION 26 23 00